

Real-Time Facial Emotion Recognition Using a Computationally Efficient CNN and Haar Cascade Face Detection

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Abstract—Facial Emotion Recognition (FER) has emerged as an important research area in artificial intelligence and computer vision due to its applications in healthcare, intelligent tutoring systems, human–computer interaction, surveillance, and driver monitoring. Although deep learning techniques have significantly improved recognition performance, many existing models require high computational resources, limiting their deployment on resource-constrained devices.

This paper presents a computationally efficient Convolutional Neural Network (CNN)-based facial emotion recognition framework integrated with Haar Cascade face detection for real-time emotion classification. The Haar Cascade classifier is employed to detect facial regions efficiently before emotion recognition, reducing computational overhead while maintaining reliable performance. The proposed CNN model classifies facial expressions into seven basic emotions: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise using the FER2013 dataset. Data preprocessing and augmentation techniques are applied to improve model robustness and generalization.

The proposed framework aims to balance recognition accuracy, computational efficiency, and inference speed, making it suitable for deployment on standard CPU-based systems and web-based applications. Experimental results demonstrate the effectiveness of the proposed approach for practical real-time facial emotion recognition.

Index Terms—Facial Emotion Recognition, Deep Learning, Convolutional Neural Network, Haar Cascade, Computer Vision, FER2013, Real-Time Emotion Recognition, Artificial Intelligence

I. INTRODUCTION

Facial Emotion Recognition (FER) is a rapidly growing research area within artificial intelligence (AI) and computer vision that enables machines to identify and interpret human emotions from facial expressions [1], [2]. Automatic recognition of emotions has become increasingly important for applications such as human–computer interaction, healthcare, intelligent tutoring systems, driver monitoring, security surveillance, and social robotics [3], [4]. Accurate emotion recognition allows intelligent systems to respond more naturally and improve user experience across diverse domains [5], [6].

Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), have significantly improved

the performance of facial emotion recognition systems [2], [5], [6]. CNNs automatically learn hierarchical facial features from images, eliminating the need for handcrafted feature extraction methods used in traditional machine learning approaches [6]. Despite these improvements, many state-of-the-art FER models require high computational resources, making them difficult to deploy on resource-constrained devices or real-time applications [2], [4].

The FER2013 dataset has become one of the most widely used benchmark datasets for evaluating facial emotion recognition systems [7]. However, it remains a challenging dataset due to its low-resolution grayscale images, variations in illumination, facial pose, occlusion, and class imbalance [2], [7]. These challenges often reduce the robustness and generalization capability of emotion recognition systems in real-world environments [4].

To address these limitations, this paper proposes a computationally efficient CNN-based facial emotion recognition framework integrated with Haar Cascade face detection. The Haar Cascade algorithm efficiently detects facial regions before classification, reducing unnecessary computations and improving inference speed. The proposed efficient CNN then classifies the detected face into one of seven universal emotion categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.

The main contributions of this work are summarized as follows:

- A computationally efficient CNN architecture designed for efficient real-time FER.
- Integration of Haar Cascade face detection to reduce computational overhead.
- Comprehensive evaluation using the FER2013 benchmark dataset.
- Development of a web-based facial emotion recognition application supporting image upload and live webcam prediction.
- A computationally efficient solution suitable for deployment on CPU-based systems.

A. Related Work

Facial Emotion Recognition (FER) has been extensively studied using both traditional machine learning and modern deep learning approaches [7]. Early FER systems relied on handcrafted feature extraction techniques combined with classifiers such as Support Vector Machines (SVMs), Random Forests, and K-Nearest Neighbors (KNN) [1], [8]. Although these methods achieved satisfactory performance on controlled datasets, their robustness under varying illumination, facial pose, and occlusion was limited [1], [8].

Recent advances in deep learning have significantly improved the performance of Facial Emotion Recognition (FER) systems through automatic feature learning [2], [4]. Convolutional Neural Networks (CNNs) have become the dominant approach because of their ability to learn hierarchical facial representations directly from images [5], [6]. Several CNN-based architectures have been proposed to achieve an effective balance between recognition accuracy and computational efficiency, making them suitable for real-time FER applications.

Recent studies have explored advanced deep learning architectures, including attention-based networks, Residual CNNs, Vision Transformers (ViTs), EfficientNet, MobileViT, and ConvNeXt, to improve facial emotion recognition performance [2], [6], [9], [10]. These models have demonstrated improved feature representation and recognition accuracy on benchmark datasets. However, many of these architectures require substantial computational resources, limiting their deployment on edge devices and real-time applications. Consequently, there remains a need for lightweight and computationally efficient FER models that maintain high recognition performance while reducing inference complexity.

Recent survey articles consistently identify several open research challenges in facial emotion recognition, including class imbalance, dataset bias, occlusion, illumination variation, cross-domain generalization, and the difficulty of distinguishing visually similar emotions such as fear and disgust [1], [2], [4]. These limitations indicate that achieving both high recognition accuracy and computational efficiency remains an active research problem, particularly for real-time and resource-constrained applications [5], [6].

B. Research Gap

Although recent deep learning models have substantially improved facial emotion recognition accuracy, several challenges remain unresolved [2], [4].

First, many high-performing FER models require significant computational resources, making them unsuitable for deployment on resource-constrained systems. Second, recognition performance decreases under real-world conditions involving illumination changes, facial occlusion, pose variation, and low-resolution images. Third, several studies report persistent confusion among visually similar emotion classes, particularly fear, disgust, and sadness. Furthermore, cross-dataset generalization remains a significant challenge because of domain shift, class imbalance, and annotation inconsistencies [1], [2].

Most existing approaches focus primarily on improving recognition accuracy while paying comparatively less attention to computational efficiency and practical deployment.

Therefore, there is a need for a computationally efficient FER framework capable of maintaining reliable recognition performance while operating in real time on standard CPU-based devices.

C. Research Objectives

The objectives of this research are summarized as follows:

- 1) Develop a computationally efficient Convolutional Neural Network for facial emotion recognition.
- 2) Integrate Haar Cascade face detection for efficient facial localization before emotion classification.
- 3) Classify facial expressions into seven universal emotion categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.
- 4) Evaluate the proposed framework using the FER2013 benchmark dataset.
- 5) Compare the proposed model with existing efficient FER approaches.
- 6) Demonstrate the feasibility of real-time deployment on CPU-based systems and web-based applications.

D. Novelty of the Proposed Work

Although several deep learning approaches have been proposed for facial emotion recognition, many existing models focus primarily on improving classification accuracy while overlooking computational efficiency and practical deployment. The proposed work addresses these aspects by integrating a custom CNN architecture with Haar Cascade face detection to develop a real-time FER system suitable for desktop and web-based applications.

The main contributions of this work are summarized as follows:

- A custom CNN architecture consisting of five convolutional layers with Batch Normalization and Dropout regularization is designed for robust seven-class facial emotion recognition.
- Haar Cascade face detection is integrated with the CNN classifier to accurately detect and crop facial regions before emotion classification, reducing unnecessary background information.
- A comprehensive data augmentation strategy, including image rotation, translation, zooming, and horizontal flipping, is employed to improve the model's generalization capability and reduce overfitting.
- An adaptive training strategy using Early Stopping, Model Checkpointing, and ReduceLROnPlateau is adopted to obtain the best-performing model while preventing unnecessary training.
- The trained model is deployed as a real-time web application using the Flask framework, allowing users to perform emotion recognition from uploaded images or live camera input without requiring specialized hardware.

- The complete system is designed to operate efficiently on CPU-based environments, making it suitable for educational, research, and efficient real-world applications.

E. Major Contributions

The major contributions of this research are as follows:

- 1) Development of a custom CNN-based facial emotion recognition model for seven emotion classes using the FER2013 dataset.
- 2) Integration of Haar Cascade face detection with deep learning-based emotion classification for efficient pre-processing.
- 3) Implementation of multiple regularization techniques, including Batch Normalization, Dropout, Early Stopping, and adaptive learning rate scheduling, to improve training stability and model generalization.
- 4) Development of a complete real-time FER web application using Python and Flask.
- 5) Comprehensive experimental evaluation using the FER2013 dataset with quantitative performance metrics and visual analysis.

II. PROPOSED METHODOLOGY

The proposed Facial Emotion Recognition (FER) framework consists of four major stages: image acquisition, face detection, image preprocessing, and emotion classification using a computationally efficient Convolutional Neural Network (CNN). Figure 1 illustrates the overall workflow of the proposed system.

The input image is captured either from a webcam or uploaded by the user through the web application. The Haar Cascade classifier first detects the facial region from the input image. The detected face is then cropped, converted to grayscale, resized to a fixed resolution of 48×48 pixels, and normalized before being passed to the CNN model. Finally, the CNN predicts one of the seven basic facial emotions.

The overall workflow is summarized as follows:

- 1) Acquire an input image from a webcam or upload.
- 2) Detect the face using the Haar Cascade classifier.
- 3) Crop and preprocess the detected facial region.
- 4) Feed the processed image into the CNN model.
- 5) Predict the facial emotion.
- 6) Display the predicted emotion to the user.

A. Overall System Architecture

The overall architecture of the proposed facial emotion recognition system is illustrated in Fig. 1. The system combines traditional computer vision techniques with a custom Convolutional Neural Network (CNN) to perform real-time FER. The workflow begins with image acquisition from either an uploaded image or a live camera feed. Haar Cascade is then employed to detect and localize the facial region. The detected face is cropped, resized to 48×48 pixels, converted to grayscale, and normalized before being passed to the CNN model. The extracted features are classified using a Softmax output layer to predict one of the seven facial emotion classes.

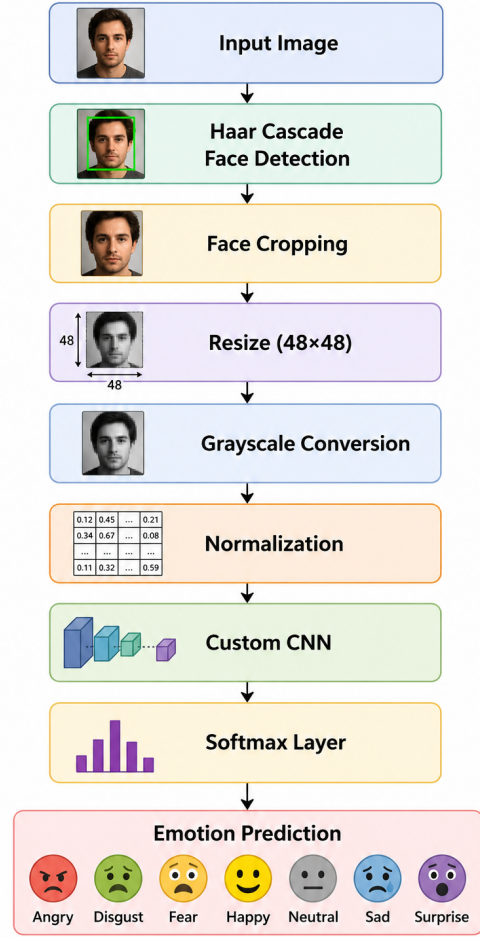


Fig. 1. Overall workflow of the proposed facial emotion recognition system.

The proposed system consists of the following stages:

- Input Image Acquisition
- Face Detection using Haar Cascade
- Face Cropping
- Image Resizing (48×48 pixels)
- Grayscale Conversion
- Image Normalization
- Feature Extraction using the Proposed CNN
- Emotion Classification using Softmax
- Web-Based User Interface

B. Dataset Description

The proposed model is trained and evaluated using the FER2013 dataset, one of the most widely used benchmark datasets for facial emotion recognition [7]. The dataset contains grayscale facial images of size 48×48 pixels collected under unconstrained real-world conditions. FER2013 consists of 35,887 facial images categorized into seven emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral [7].

The dataset is divided into training, validation, and testing subsets to evaluate the generalization capability of the

proposed model. The official FER2013 split contains 28,709 training images, 3,589 public test (validation) images, and 3,589 private test images, enabling consistent performance evaluation across different facial emotion recognition models [7].

C. Data Preprocessing

To improve the robustness of the CNN model, several preprocessing operations are applied before training.

The preprocessing pipeline includes:

- Face detection using Haar Cascade.
- Conversion of RGB images into grayscale.
- Resizing images to 48×48 pixels.
- Pixel normalization into the range $[0,1]$.
- Data augmentation using random rotation, horizontal flipping, zooming, and shifting.

These preprocessing techniques reduce overfitting and improve the model's ability to generalize to unseen facial images.

D. Face Detection Using Haar Cascade

Face detection is performed using the Viola–Jones Haar Cascade classifier implemented in OpenCV [11], [12]. The Haar Cascade algorithm detects facial regions by evaluating Haar-like features using an AdaBoost-based cascade of classifiers, enabling rapid and efficient object detection [11].

Compared with deep learning-based face detectors, the Haar Cascade classifier provides fast inference with low computational complexity, making it suitable for real-time applications running on CPU-based devices [11]. Only the detected facial region is forwarded to the CNN classifier, thereby reducing unnecessary computations and improving inference speed.

E. CNN Architecture

The proposed facial emotion recognition system employs a custom Convolutional Neural Network (CNN) designed to classify facial expressions into seven emotion categories. The network accepts grayscale facial images of size 48×48 pixels as input and performs hierarchical feature extraction through multiple convolutional layers followed by fully connected layers for classification.

The architecture consists of five convolutional layers with 32, 32, 64, 64, and 128 filters, respectively. Each convolutional layer uses a 3×3 kernel and the Rectified Linear Unit (ReLU) activation function to learn discriminative facial features. Batch Normalization layers are incorporated after selected convolutional layers to improve training stability and accelerate convergence. Max-Pooling layers with a pool size of 2×2 reduce the spatial dimensions of the feature maps, while Dropout layers are employed after each convolutional block to minimize overfitting and improve the generalization capability of the network.

The extracted feature maps are flattened and passed to a fully connected layer containing 512 neurons with ReLU activation. A Dropout layer with a rate of 0.50 is applied before the final classification layer. The output layer consists of seven neurons with a Softmax activation function, which

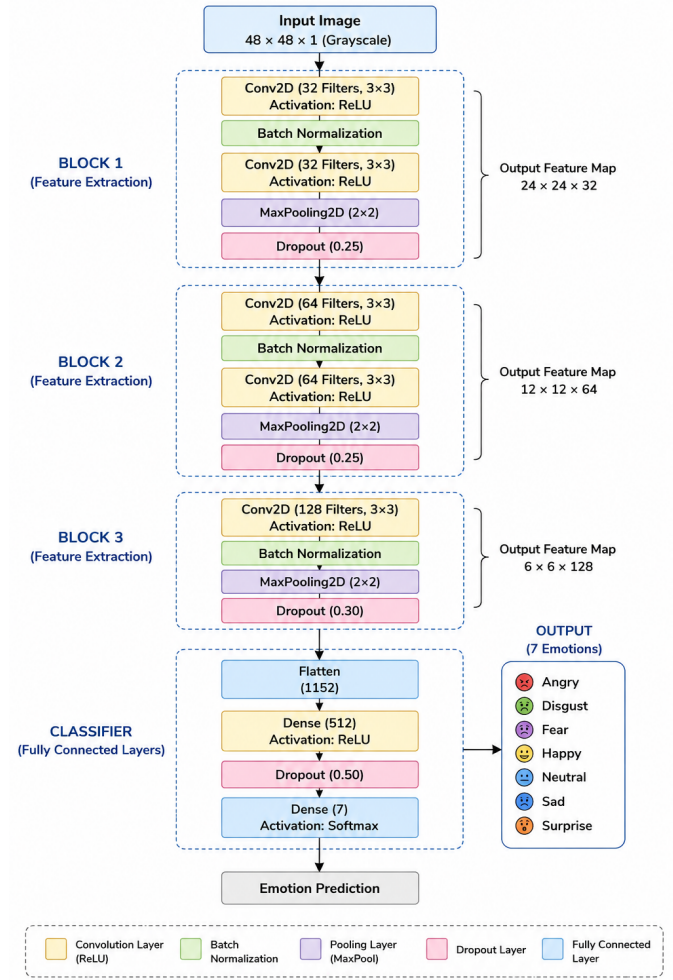


Fig. 2. Architecture of the proposed CNN model for facial emotion recognition.

produces the probability distribution over the seven facial emotion classes.

Figure 2 illustrates the architecture of the proposed CNN model used for facial emotion recognition.

Table I summarizes the configuration of each layer in the proposed CNN architecture.

The proposed CNN is designed to achieve a balance between recognition performance and computational efficiency. According to the model summary, the network contains 733,223 trainable parameters and 448 non-trainable parameters associated with the Batch Normalization layers. The serialized model occupies approximately 8.39 MB of storage, making it suitable for deployment in desktop and web-based facial emotion recognition applications while maintaining moderate computational complexity.

III. MODEL TRAINING AND EXPERIMENTAL SETUP

A. Training Procedure

The proposed CNN model was trained using the FER2013 dataset. Before training, all facial images were preprocessed

TABLE I
ARCHITECTURE OF THE PROPOSED CNN MODEL

Layer	Configuration
Input	$48 \times 48 \times 1$ (Grayscale)
Conv2D	32 filters, 3×3 , ReLU
Batch Normalization	Yes
Conv2D	32 filters, 3×3 , ReLU
MaxPooling2D	2×2
Dropout	0.25
Conv2D	64 filters, 3×3 , ReLU
Batch Normalization	Yes
Conv2D	64 filters, 3×3 , ReLU
MaxPooling2D	2×2
Dropout	0.25
Conv2D	128 filters, 3×3 , ReLU
Batch Normalization	Yes
MaxPooling2D	2×2
Dropout	0.30
Flatten	1152 Features
Dense	512 Neurons, ReLU
Dropout	0.50
Output	7 Neurons, Softmax

by converting them to grayscale, resizing them to 48×48 pixels, and normalizing pixel values to the range of 0–1. Data augmentation techniques, including horizontal flipping, random rotation, zooming, and width/height shifting, were applied to increase the diversity of the training data and reduce overfitting.

The model was trained using mini-batch gradient descent. During each training epoch, batches of augmented images were forwarded through the network, and the model parameters were updated using backpropagation. Early stopping and model checkpointing were employed to prevent overfitting and retain the best-performing model based on validation accuracy.

B. Hyperparameter Settings

Table II summarizes the hyperparameters and training configuration used for the proposed CNN model.

C. Loss Function

Since facial emotion recognition is a multi-class classification problem, the Categorical Crossentropy loss function was used during training. The objective is to minimize the difference between the predicted probability distribution and the true class labels.

The categorical crossentropy loss is defined as

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (1)$$

where C represents the total number of emotion classes, y_i denotes the true class label, and $\hat{y}_i$ represents the predicted probability for class i .

D. Optimizer

The Adam optimizer was employed because of its fast convergence and adaptive learning capability. Adam combines the advantages of momentum and adaptive learning rate optimization, making it one of the most widely used optimization algorithms for deep learning applications.

TABLE II
TRAINING HYPERPARAMETERS OF THE PROPOSED CNN MODEL

Parameter	Value
Dataset	FER2013
Training Images	28,709
Testing Images	7,178
Input Image Size	48×48 pixels
Color Mode	Grayscale
Batch Size	64
Maximum Epochs	50
Optimizer	Adam
Learning Rate	0.001 (Default Adam)
Loss Function	Categorical Crossentropy
Output Classes	7 (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise)
Hidden Layer Activation	ReLU
Output Layer Activation	Softmax
Dropout Rates	0.25, 0.25, 0.30, 0.50
Batch Normalization	Applied after convolutional layers
Data Augmentation	Rotation (15°), Width Shift (10%), Height Shift (10%), Zoom (20%), Horizontal Flip
Early Stopping	Monitor: Validation Loss, Patience = 8
Learning Rate Scheduler	ReduceLROnPlateau (Factor = 0.2, Patience = 3)
Model Checkpoint	Save Best Model based on Validation Accuracy
Evaluation Metric	Accuracy

The parameter update rule is expressed as

$$\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (2)$$

where θ represents the model parameters, α is the learning rate, $\hat{m}_t$ and $\hat{v}_t$ denote the bias-corrected first and second moment estimates, and ϵ is a small constant used for numerical stability.

E. Evaluation Metrics

The performance of the proposed model was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Accuracy is calculated as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Precision, Recall, and F1-score provide additional insight into the classification performance for individual emotion classes, especially in the presence of class imbalance.

F. Experimental Environment

The proposed framework was implemented using Python with TensorFlow and Keras. OpenCV was used for face detection and image processing. Model training was performed on a Windows-based system equipped with an Intel Core

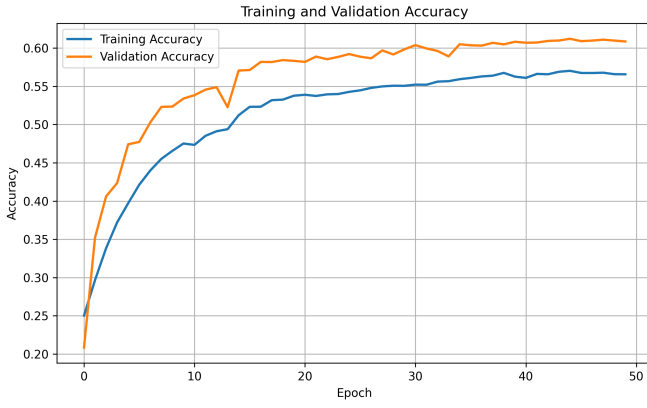


Fig. 3. Training and validation accuracy of the proposed CNN model on the FER2013 dataset.

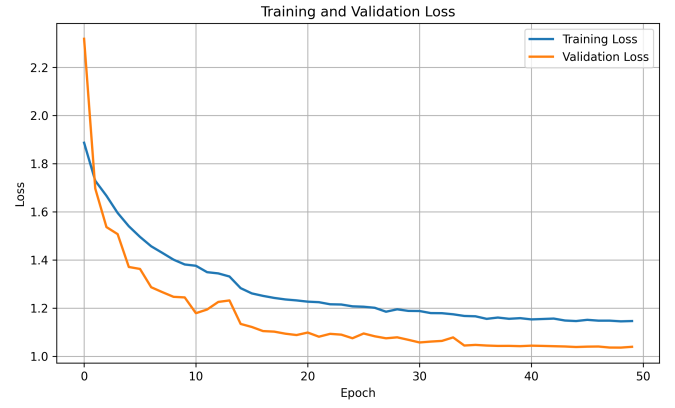


Fig. 4. Training and validation loss of the proposed CNN model during training.

processor and sufficient RAM for deep learning experimentation. The trained model was integrated into a Flask-based web application for real-time FER using uploaded images and webcam input.

IV. RESULTS AND DISCUSSION

The proposed CNN model was trained for 50 epochs on the FER2013 dataset using the Adam optimizer with data augmentation and regularization techniques. The model achieved a final test accuracy of **60.98%** with a corresponding test loss of **1.0358**. The highest validation accuracy obtained during training was approximately **61.21%**.

Figures 3 and 4 illustrate the training and validation accuracy and loss curves, respectively. The accuracy increased steadily throughout the training process and stabilized after approximately 30 epochs, while both training and validation losses decreased consistently, indicating successful convergence of the proposed model.

The validation accuracy remained slightly higher than the training accuracy because dropout and data augmentation were applied during training, whereas validation was performed on unmodified images. The small gap between the training and validation curves indicates that the proposed model generalizes well without exhibiting significant overfitting.

Figure 5 presents the confusion matrix of the proposed model. The model achieved the highest recognition performance for the *Happy* and *Surprise* emotion classes, while *Fear* and *Disgust* were the most challenging categories. The confusion matrix also shows that visually similar emotions such as *Neutral* and *Sad* were occasionally misclassified, which is consistent with the challenging nature of the FER2013 dataset.

Overall, the proposed CNN demonstrates reliable facial emotion recognition performance while maintaining relatively low computational complexity, making it suitable for real-time FER applications on standard CPU-based systems.

A. Confusion Matrix Analysis

Figure 5 presents the confusion matrix of the proposed CNN model evaluated on the FER2013 test dataset. The results show

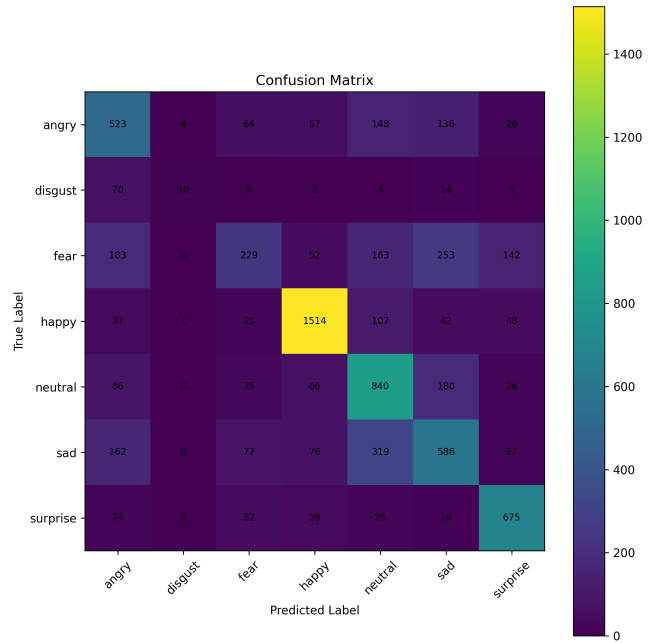


Fig. 5. Confusion matrix of the proposed CNN model evaluated on the FER2013 test dataset.

that the model achieved the highest recognition performance for the *Happy* and *Surprise* emotion classes, with F1-scores of 0.85 and 0.76, respectively. In contrast, the *Fear* and *Disgust* classes exhibited comparatively lower recognition performance because of their visual similarity to other emotions and the limited number of training samples available in the dataset.

The confusion matrix further indicates that the model occasionally confused *Neutral* with *Sad* and *Fear* with *Angry* or *Sad*, which is consistent with the challenging characteristics of the FER2013 dataset. Despite these challenges, the proposed CNN successfully learned discriminative facial features and achieved an overall classification accuracy of 60.98%, demonstrating its effectiveness for real-time FER on standard CPU-based systems.

TABLE III
COMPARISON OF THE PROPOSED CNN WITH EXISTING FER METHODS
ON THE FER2013 DATASET

Method	Accuracy (%)	Parameters (M)
MobileViT (Base) [13]	61.00	5.60
Improved MobileViT [14]	62.20	0.95
Improved MobileNetV2 [15]	68.62	3.50
EfficientNet-B0 [10]	69.85	5.30
Proposed CNN	60.98	0.73

TABLE IV
CLASSIFICATION PERFORMANCE OF THE PROPOSED CNN

Emotion	Precision	Recall	F1-score
Angry	0.48	0.55	0.51
Disgust	0.59	0.09	0.16
Fear	0.47	0.22	0.30
Happy	0.84	0.85	0.85
Neutral	0.52	0.68	0.59
Sad	0.48	0.47	0.47
Surprise	0.71	0.81	0.76

B. Comparison with Existing Methods

Table III compares the proposed CNN with several representative facial emotion recognition methods evaluated on the FER2013 dataset. The results indicate that although advanced lightweight CNNs and transformer-based architectures achieve higher recognition accuracy, they generally require more sophisticated network designs and increased computational resources. The proposed CNN achieves competitive performance while maintaining a simple architecture suitable for real-time deployment on standard CPU-based systems.

The proposed CNN achieved an overall accuracy of 60.98% on the FER2013 dataset. Although recent lightweight architectures such as MobileNetV2, EfficientNet-B0, and Improved MobileViT report higher recognition accuracy, these models employ more sophisticated network designs and increased computational complexity. In contrast, the proposed CNN combines a lightweight convolutional architecture with Haar Cascade face detection, providing an effective balance between recognition performance, computational efficiency, and ease of deployment on standard CPU-based systems and web-based applications.

C. Classification Performance

Table IV presents the precision, recall, and F1-score obtained for each emotion class on the FER2013 test dataset. The proposed CNN achieved the highest performance for the *Happy* and *Surprise* classes, while lower recognition rates were observed for *Fear* and *Disgust*. These results are consistent with the challenging characteristics of the FER2013 dataset, where visually similar expressions and class imbalance make accurate classification more difficult.

D. Discussion

The proposed computationally efficient CNN successfully performs real-time FER while maintaining a simple architecture suitable for deployment on standard CPU-based systems.

The integration of Haar Cascade face detection reduces computational overhead by extracting only the facial region before classification.

Although the proposed model demonstrates reliable real-time performance, recognition accuracy remains affected by several factors, including low image resolution, illumination variations, facial occlusions, and the inherent class imbalance of the FER2013 dataset. Emotions such as Fear and Disgust continue to be more challenging to classify due to their subtle facial characteristics.

Future improvements may include transfer learning, attention mechanisms, improved data augmentation strategies, temporal modeling for video-based emotion recognition, and evaluation on additional benchmark datasets.

E. Limitations

Although the proposed CNN achieves reliable performance on the FER2013 dataset, recognition accuracy remains limited for visually similar emotions such as Fear and Disgust. Future work may explore attention mechanisms, transformer-based architectures, multimodal emotion recognition, and larger datasets to improve robustness.

V. CONCLUSION

This work demonstrates that combining Haar Cascade face detection with a lightweight CNN provides an effective solution for real-time FER. The proposed framework achieves competitive recognition performance while maintaining low computational complexity, making it suitable for deployment on standard CPU-based systems and web-based applications.

Experimental evaluation demonstrated that the proposed framework can perform real-time FER using a computationally efficient architecture suitable for deployment on CPU-based systems and web-based applications.

Future work will focus on improving recognition accuracy through transfer learning, attention-based architectures, larger and more diverse datasets, and multimodal emotion recognition.

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